

Cesar Briones

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Experience

Society of Aeronautics and Rocketry (USF Rocketry Team)

Tampa, FL

Chief Engineer – Rocket Engine Development

May 2026 – Present

- Led development of a 40 bar 2.9kN rocket engine made out of steel and aluminum, fueled by ethanol and nitrous oxide
- Designed P&ID of a pressure fed engine detailing component pressures for water flow, cold flow and static fire configurations
- Performed CFD in ANSYS Fluent to extract preliminary discharge coefficients from mass flow predictions and spray cone angle
- Performed transient thermal simulation to assess structural integrity of heat sink nozzle exposed to 3000K gases for 3 seconds
- Developed remotely actuated ball valve by modifying a COTS valve, using an 80kg servo and a custom CNC aluminum holder
- Assembled Swagelok tubing, built piping support using aluminum extrusions and designed rig to test solenoids, pneumatic valves, servo valves, thermocouples and pressure transducers

Chief Engineer – IREC 10K Student Competition

May 2025 – June 2026

- Responsible for technical development, systems engineering and integration of 7 subsystems in student-built rocket
- Supervised all manufacturing operations of 25+ aluminum components via CNC programming, vertical mill operations, drill press work, and tapping, ensuring dimensional accuracy, resulting in the year with most in house made parts
- Conducted system design reviews to confirm readiness, coordinating with faculty mentors and competition judges
- Established top-level mission requirements and design constraints through trade studies and competition rule analysis
- Mentored 12+ junior members in CAD/CAM, CNC machining, FEA, CFD, and systems engineering principles

Airbrakes Lead – NASA Student Launch Competition

Jun 2024 – May 2025

- Executed parametric CFD simulations in ANSYS Fluent across deployment angles (0-100%) and Mach regimes (0.3 - 0.7), characterizing drag force profiles to support flight control algorithm development geometry and torque
- Designed a four-bar linkage mechanism for drag control, performing kinematic analysis to optimize deployment
- Developed a regression model in MATLAB from CFD, enabling fast drag lookup for the PID control system
- Validated CFD-predicted force through experimental flight testing, comparing data against simulation results
- Achieved 26% mass reduction in CNC-machined aluminum components through iterative FEA-driven design optimization while maintaining structural safety factors, allowing to stay within the team derived mass requirement.

USF Corrosion Research Laboratory

Tampa, FL

Instrumentation Researcher – Corrosion prediction in tendons

May 2024 – Dec 2025

- Designed a 2-layer PCB using EasyEDA and SMD components, reducing assembly complexity and cost replication
- Developed FFT-based signal processing algorithms in Python to detect voids in post-tensioned tendons by comparing measured impedance against reconstructed baseline signatures, implementing filtering and polynomial correction
- Developed a real-time wireless data acquisition pipeline with Arduino-to-PC serial communication (UART) at 115200 baud, processing extracted sensor data in Python with automated anomaly detection and live visualization
- Designed 3D-printed enclosures and sensor mounting hardware for portable field deployment, enabling rapid prototyping iterations and tool-free assembly for bridge structure inspections

RANCS Research Laboratory

Tampa, FL

Research Lab Assistant – Autonomous Systems

Dec 2023 – Mar 2024

- Designed and built a t-slot aluminum frame to elevate a \$10,000 LIDAR sensor, improving field of view and eliminating interferences with vehicle roof, enabling 100+ hours of successful autonomous vehicle testing
- Coordinated with PI and graduate students to define LIDAR mounting requirements, designing a cost-effective solution using existing lab inventory of t-slot aluminum extrusions and hardware

Education

University of South Florida – **M.S. in Aerospace Engineering**

Aug 2026 - Dec 2027

University of South Florida – **B.S. in Mechanical Engineering**

Aug 2022 - May 2026

Key Courses: Aerodynamics, Propulsion, Vibrations, Finite Element Methods

GPA: 3.72/4.0

Skills

CAD/Coding:

Siemens NX, SolidWorks, Python, MATLAB, Simulink, C#, LaTeX, Epsilon3

Design:

GD&T, DFMA, P&ID, Prototyping, CNC Milling Programming and Operation, Soldering

Analysis Tools:

ANSYS Workbench / Mechanical / Fluent, Excel, LabVIEW, FMEA, Hazard Analysis, Requirements

Certifications:

CSWP Simulation, CSWA Design, CSWA Additive Manufacturing

Fluids:

Swagelok, P&ID, Pneumatics, Hydraulics, Valves, Regulators